

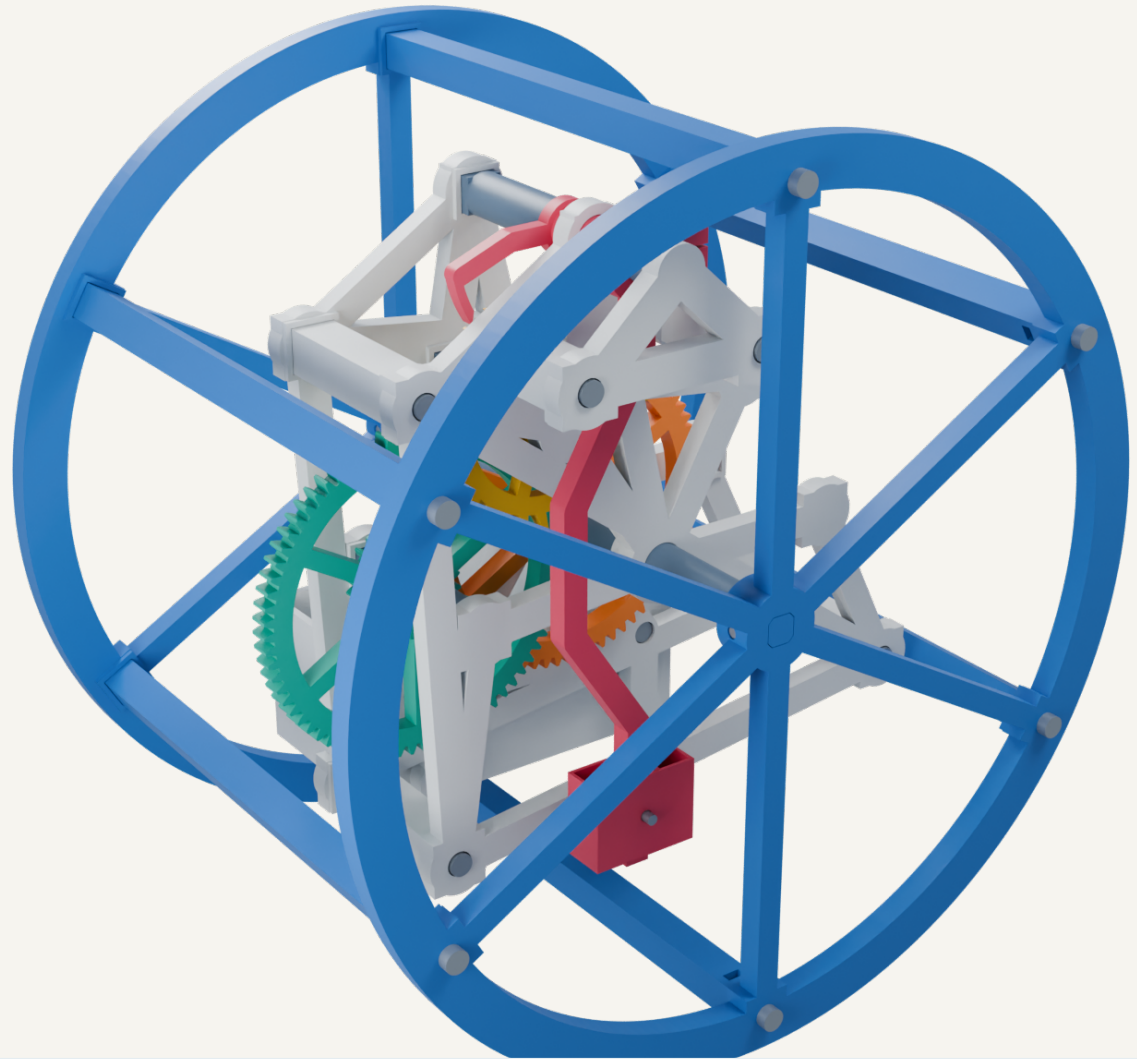
Build the bolted rolling clock

A rigid frame, a wider pendulum compartment, and three gear ratios.

Start with 16:1.

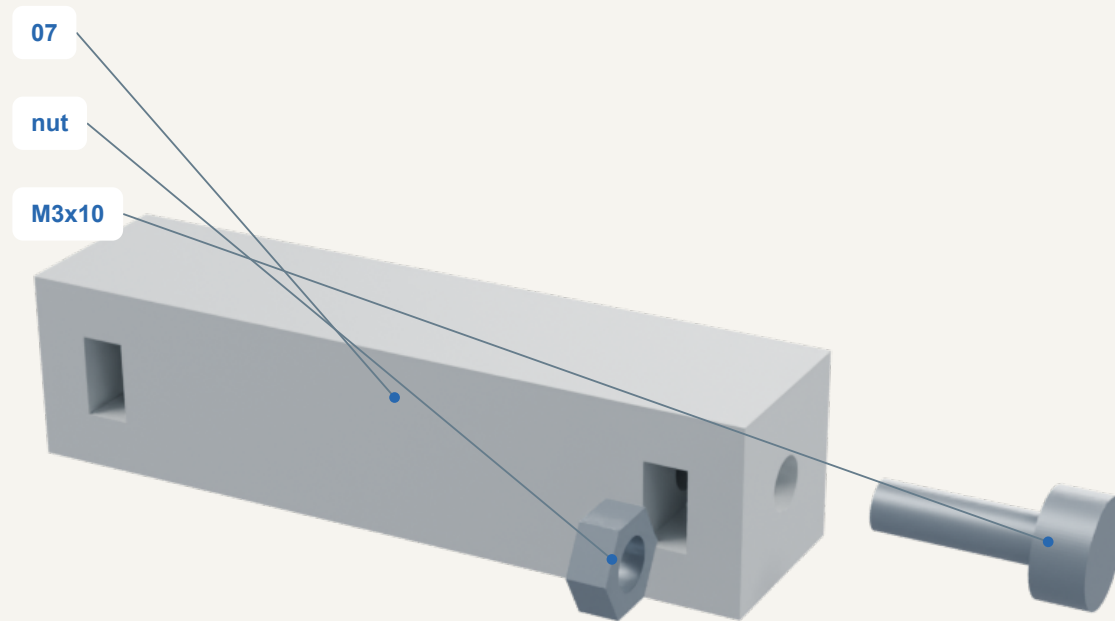
The same frames also accept 32:1 and 64:1. This guide shows the complete assembly, the captive-nut joints and the parts to change for slower rolling.

Status: the earlier lightweight clock ran successfully. R3 has passed CAD clearance checks and has not yet been printed or run. Keep the original working clock intact until this revision is proven.



Use **PRINT-LIST-16x.md** for exact filenames and quantities. For the other ratios use their own lists. All numbered parts in this guide are R3 parts; earlier revisions are not interchangeable.

Check the nut fit before the large prints



Print the small pair first: 40_nut_post_coupon_525 and 41_post_seat_coupon from STL/fit-tests/.

Slide a real M3 nut into the **5.25 mm slot**. It must sit flat on the screw axis. The post should slip into the 12.4 mm lip opening without spreading it.

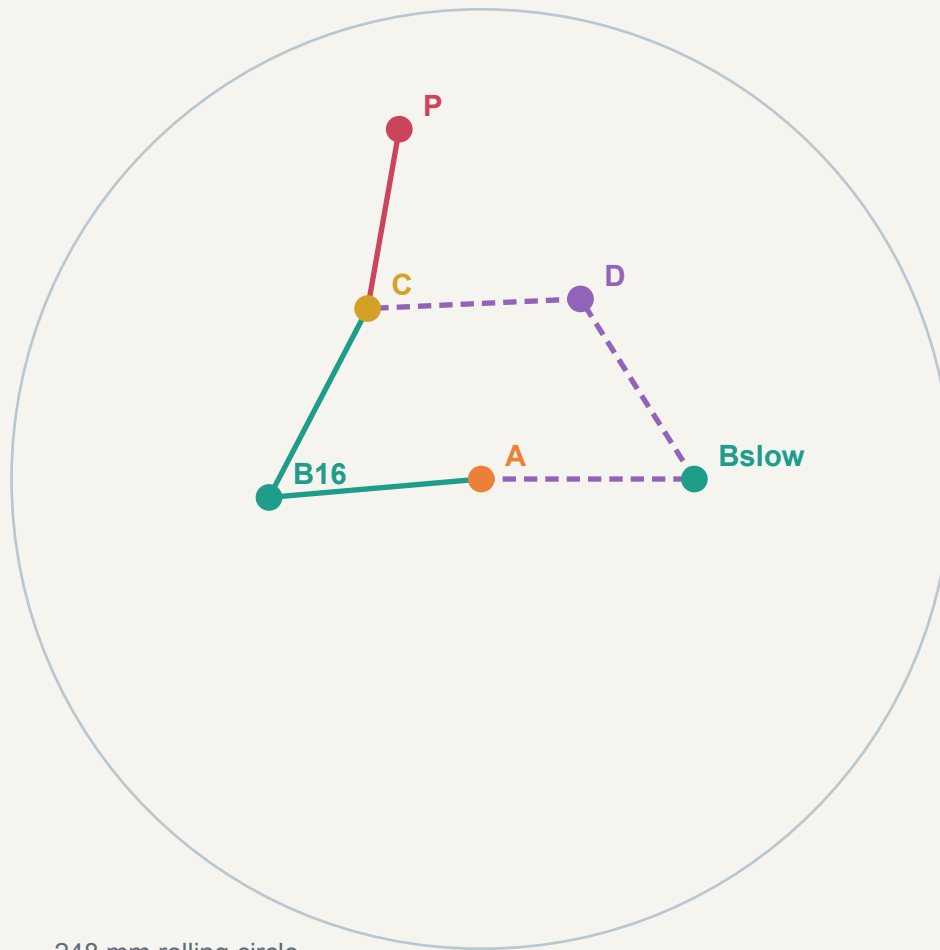
Use one **M3x10** screw through the seat coupon. Tighten until the flat faces meet. Test 5.40 or 5.60 mm alternatives if needed; adjust NUT_WIDTH and regenerate rather than scaling the whole design.

16x: 15 M3x10, 17 M3x16, 32 nuts.

32x / 64x: 16 M3x10, 17 M3x16, 33 nuts.

Square posts print on a long flat side, with nut slots facing upward. Preload both end nuts before assembly. Screw heads: flat underside, at most 6.5 mm diameter x 3.2 mm high. Plain nuts: about 2.4 mm thick.

One pair of frames, three layouts



248 mm rolling circle

A is the square rolling axle. **C** is the escape-wheel pivot. **P** is the long anchor pivot. These stay fixed for every ratio.

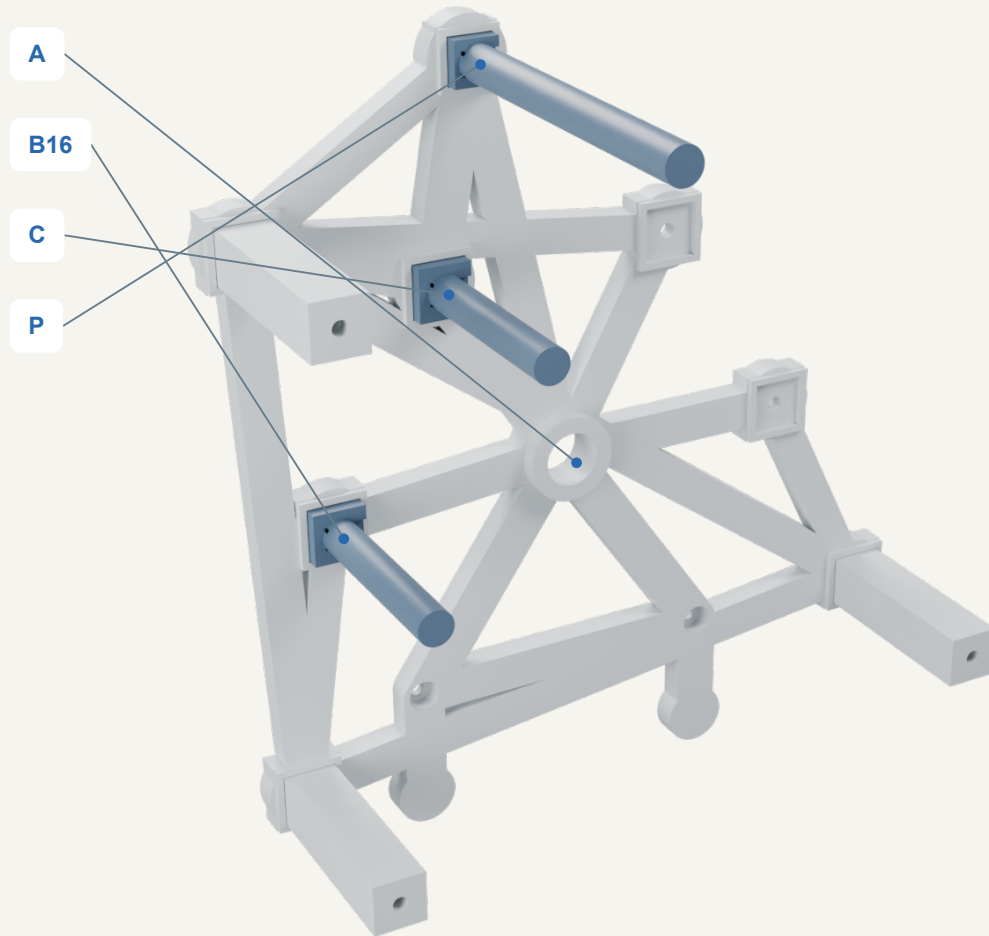
16x: fit one short pivot at B16 and one at C. Leave Bslow and D empty.

32x or 64x: fit short pivots at Bslow, D and C. Leave B16 empty. Both slower layouts use the same shaft positions.

Each pivot foot sits inside its loose square locating lip on rear frame 05. The front frame has matching **blind seats** for the gear-pivot tips. The large top opening passes the anchor sleeve.

The extra holes are deliberate. Install only the pivots for your chosen ratio. Match the frame to this view before tightening; unused seats require no plugs or screws.

Bolt the rear frame and pivot feet



1. Lay **05** with its collars and locating lips facing you. The ballast-bowl bracket is at the bottom.

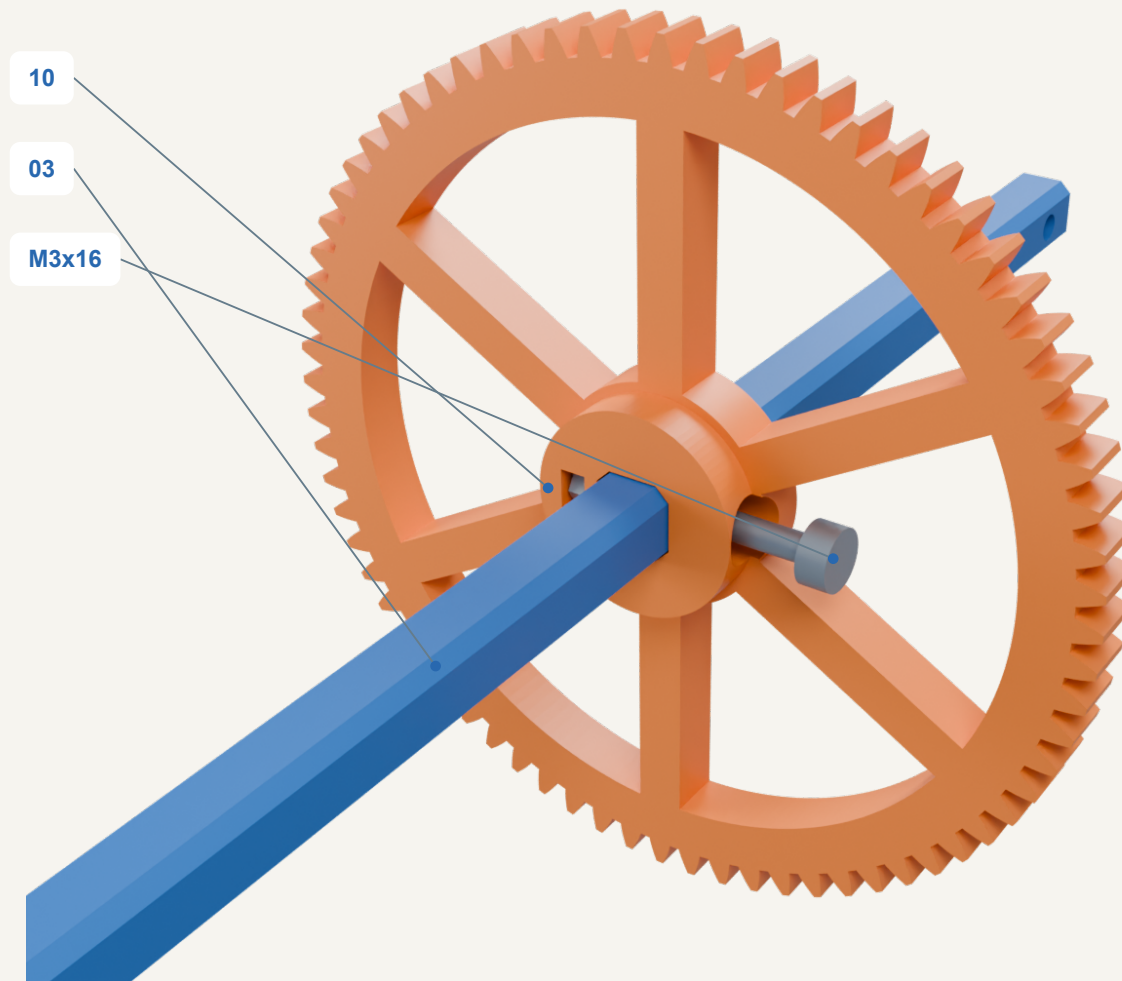
2. Preload the nuts. Seat the two **13** gear pivots at **B16** and **C**, and the long **19** pivot at P. Fasten each from behind with **M3x10**.

3. Fit the three **07** square frame posts inside their locating lips. Use one **M3x10** from behind for each post; leave the front nuts loaded.

4. Bring the flat shoulders into contact. Do not use the screws to pull a crooked pivot or a tight locating lip into place.

The front frame will be 48 mm from the rear frame's inside face, set by the flat-ended posts. The frame-post nuts must be fully seated. Gear journals remain printed, smooth, clearance-fit surfaces.

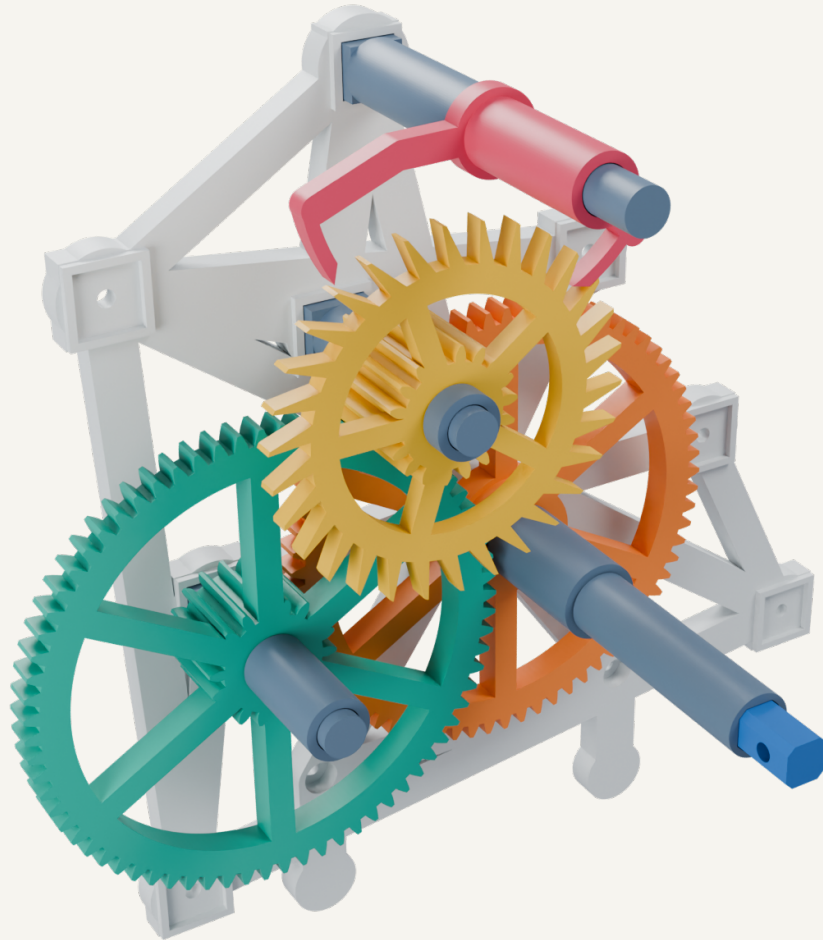
Fit the square axle and orange gear



1. Identify and mark the rear end of **03**: the middle cross-hole is **91 mm** from that end (89 mm from the other). The end holes are each 6 mm from their nearest end.
2. Slide long rear journal **36 (70 mm)** onto the axle and through A. Add rear thrust washer **35 (1.2 mm)**.
3. Preload a nut into the orange gear's hub slot. Slide on **10**, tall hub forward, align the middle cross-hole and fit an **M3x16** screw. Its head sits in the circular side recess.
4. Add keyed front thrust sleeve **34 (26.4 mm)** ahead of the orange hub. The square faces carry torque; the cross-screw holds the gear in position.

Main stack, rear to front: rear wheel hub -> 36 -> 35 -> orange gear 10 -> 34 -> front journal 37 -> front wheel hub. Wheel hubs are fitted later. Keep loose sleeves from sliding off during the bench build.

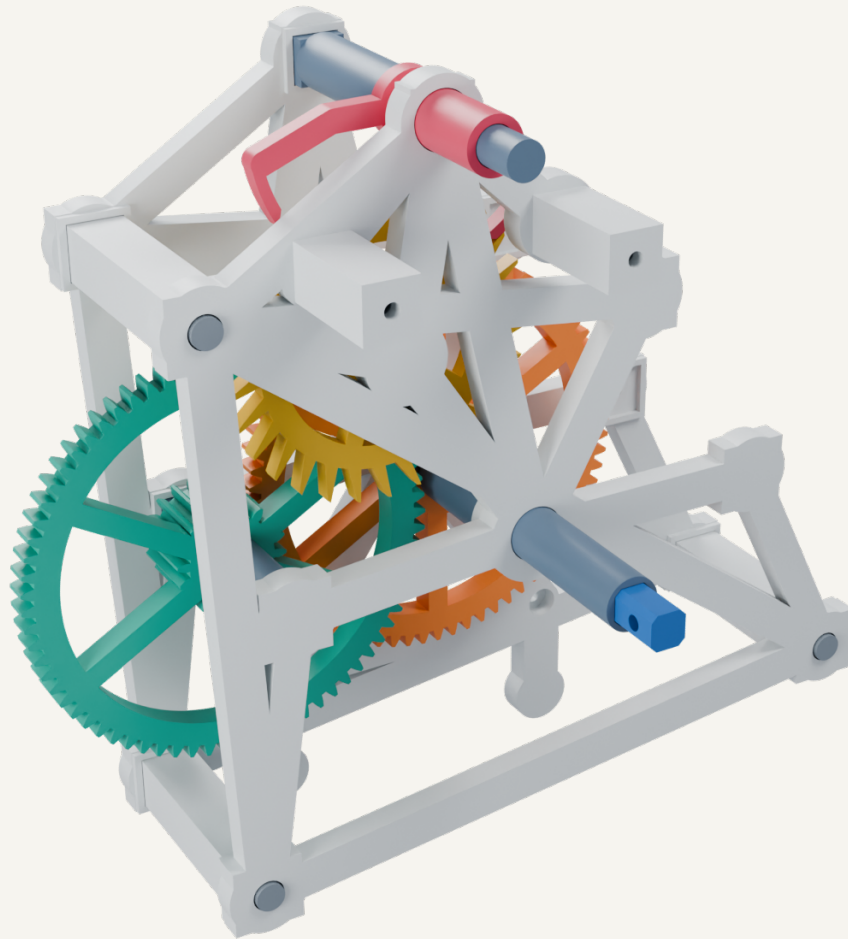
Build the two gear stages



1. On the B16 pivot: add **15 (1.2 mm)**, then green **11** with its small pinion rearward, then **16 (23.7 mm)**. The small pinion engages the orange wheel.
2. On C: add **17 (13.2 mm)**, then yellow **12** with its small pinion rearward, then **27 (4.7 mm)**. Its small pinion engages the large green wheel.
3. On P: add **20 (34.2 mm)**, then the **18_anchor_forward** from STL/16x/. Its long sleeve points forward. Ease the pallet faces around the escape-wheel teeth.
4. Turn the input gently while supporting the pivots. Stop at a locked pallet; never force the wheel past it.

All gear spacers have flat annular ends. The intended endplay is 0.6 mm per gear. Neither the front plate nor a screw should clamp a rotating gear tightly between fixed faces.

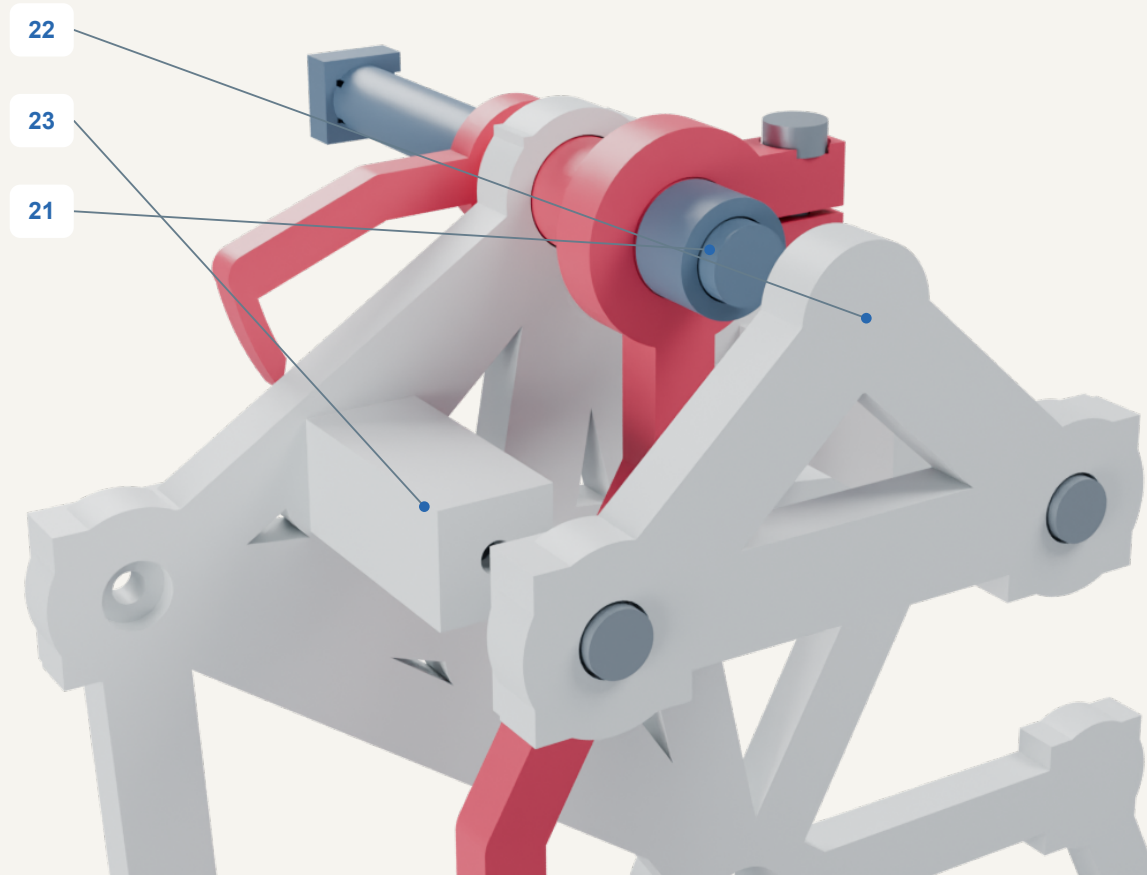
Prepare and close the front frame



1. Before fitting **06**, bolt both **23** bridge posts to its outer face with **M3x10** screws inserted from the gear side. Their ends sit flat against the frame.
2. Temporarily use bridge **22** as an alignment template over the two posts while tightening their rear screws. Its square lips prevent twisting. Lift the bridge off again.
3. Place frame 06 over the three frame posts, both short pivot tips and the anchor sleeve. Its blind bearing seats and A collar face inward, toward the gears.
4. Add the three front **M3x10** frame-post screws, seating them gradually. Slide short front journal **37 (45.6 mm)** onto the axle and into A.

If anything binds as the frame closes, reopen it and check the spacer order and blind seats. There should be 0.5 mm behind each seated pivot tip. Do not tighten a screw to overcome a misalignment.

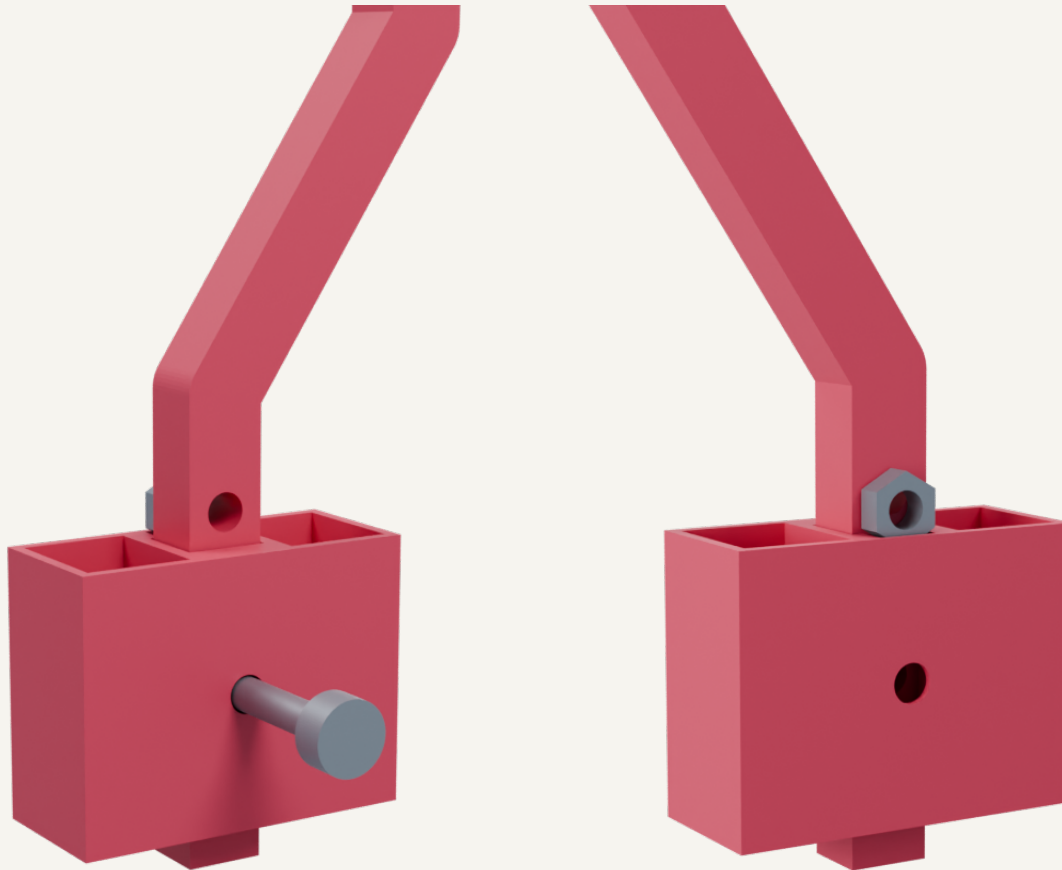
Clamp the pendulum and close the bridge



1. Slide the head of **24** onto the anchor sleeve, ahead of the frame. Its bent rod runs down around the main axle. Keep the front of the clamp approximately flush with the sleeve end.
2. Fit the clamp nut and **M3x16** screw. The screw head sits in the shallow circular recess. Tighten lightly so the angle can still be adjusted for the beat.
3. Place **21 (7.7 mm)** on the exposed anchor pivot, ahead of the sleeve and pendulum head.
4. Fit bridge **22** onto both posts and the pivot tip. Add its two front **M3x10** screws and snug them evenly. Confirm that the pendulum swings freely.

The bridge locates the fixed pivot; it must not squeeze the rotating sleeve. The square posts and flat faces establish the spacing. Check clearance at both clamp ears and the screw through a gentle swing.

Fasten the bob with a real screw



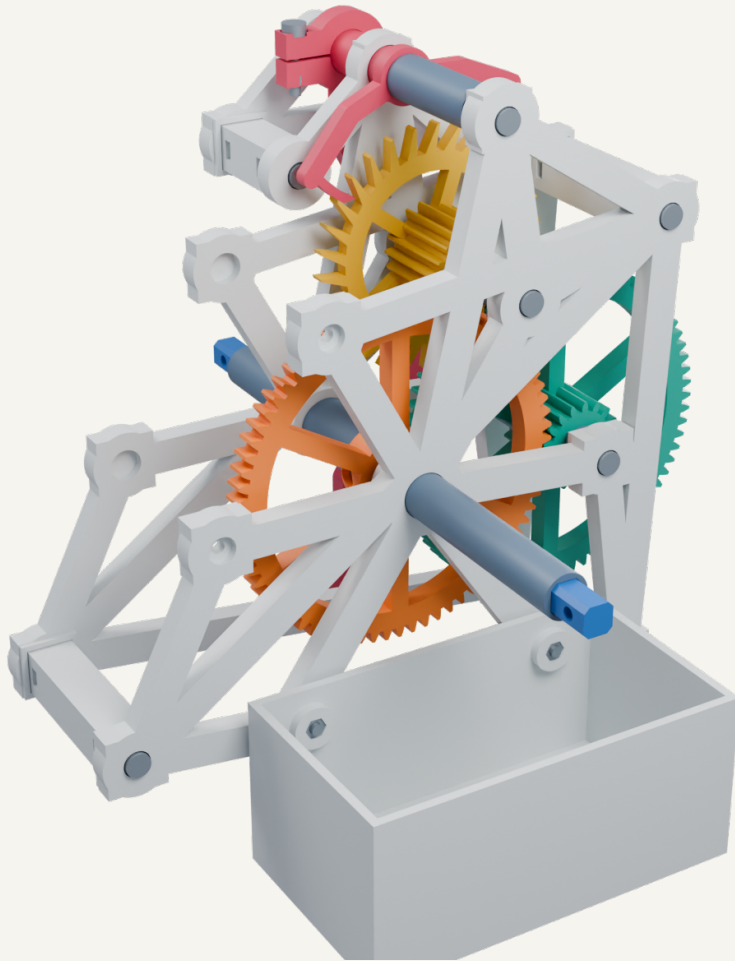
Rear: insert the screw

Front: drop in the nut

1. Slide bob **25** over the straight lower part of the rod, with both iron pockets facing upward. Start at the **150 mm hole**: fourth from the pivot, second from the free end.
2. Drop a plain M3 nut into the central slot on the **front side** of the bob. This slot is separate from the two iron pockets. Align its thread with the rod hole.
3. Insert one **M3x16** from the rear, toward the front nut. Tighten until the bob is held; do not crush its thin outside walls.
4. With the frame upright, add **15-20 g total** iron, approximately evenly divided. Keep the main ballast bowl empty until the drum is fitted.

Nominal clearances: bob to frame 9.4 mm; screw head to frame 6.2 mm; screw tip to front spokes 15.6 mm. The 1 mm outer walls save space; the central web and thicker floor carry the attachment and ballast loads.

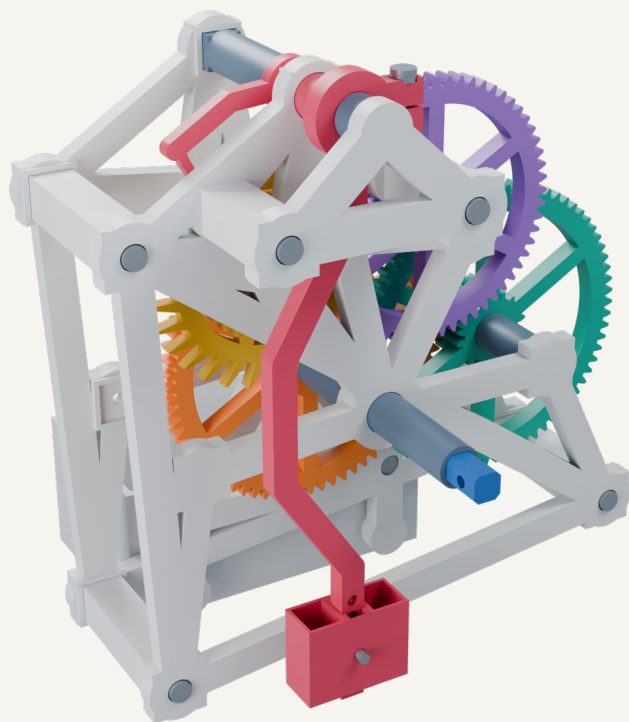
Bench-test, then add bowl and drum



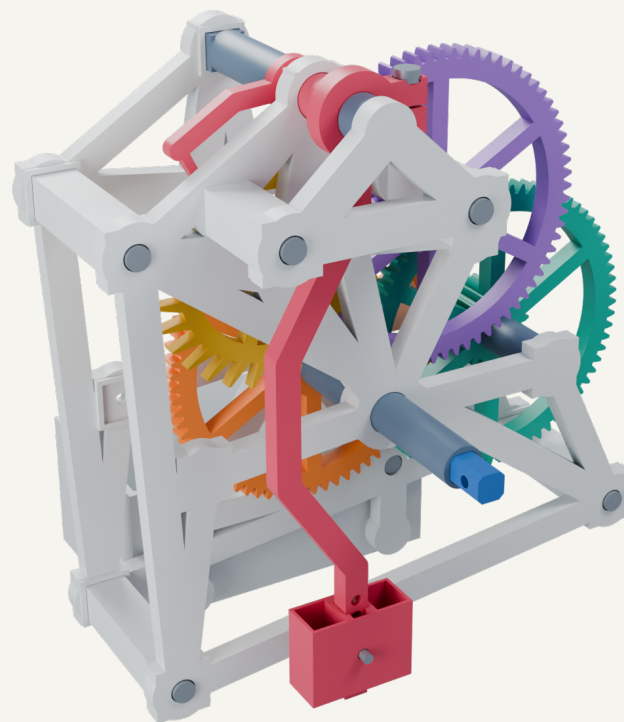
1. Support the fixed frame upright, without squeezing its bearings. Keep the loose axle sleeves seated with removable tape collars if necessary. Apply light **counterclockwise** input torque, seen from the pendulum side.
2. Adjust the pendulum clamp angle until each half-swing releases one step and catches on the other pallet. Then snug the clamp. Remove temporary tape before final assembly.
3. Fit empty bowl **08** with two **M3x10** screws from the gear side and nuts inside the cup. Both lower pads must touch the rear frame.
4. Preload nuts in both wheel hubs and all six case posts. Fit the rear wheel and its **M3x16 axle screw**; bolt the six posts to that wheel with six M3x16. Fit the front wheel, its axle screw, and six front post screws.

The case uses 12 post screws plus 2 axle screws. Snug opposite posts progressively with every shoulder seated. Turn the case through a full revolution while gently moving the pendulum before adding about 600 g to the bowl.

Swap gearing without reprinting the frames



32x: final pair 60:30



64x: final pair 72:18

16x to a slower set: move B's pivot to Bslow. Add a third short pivot at D and its M3x10 screw/nut.

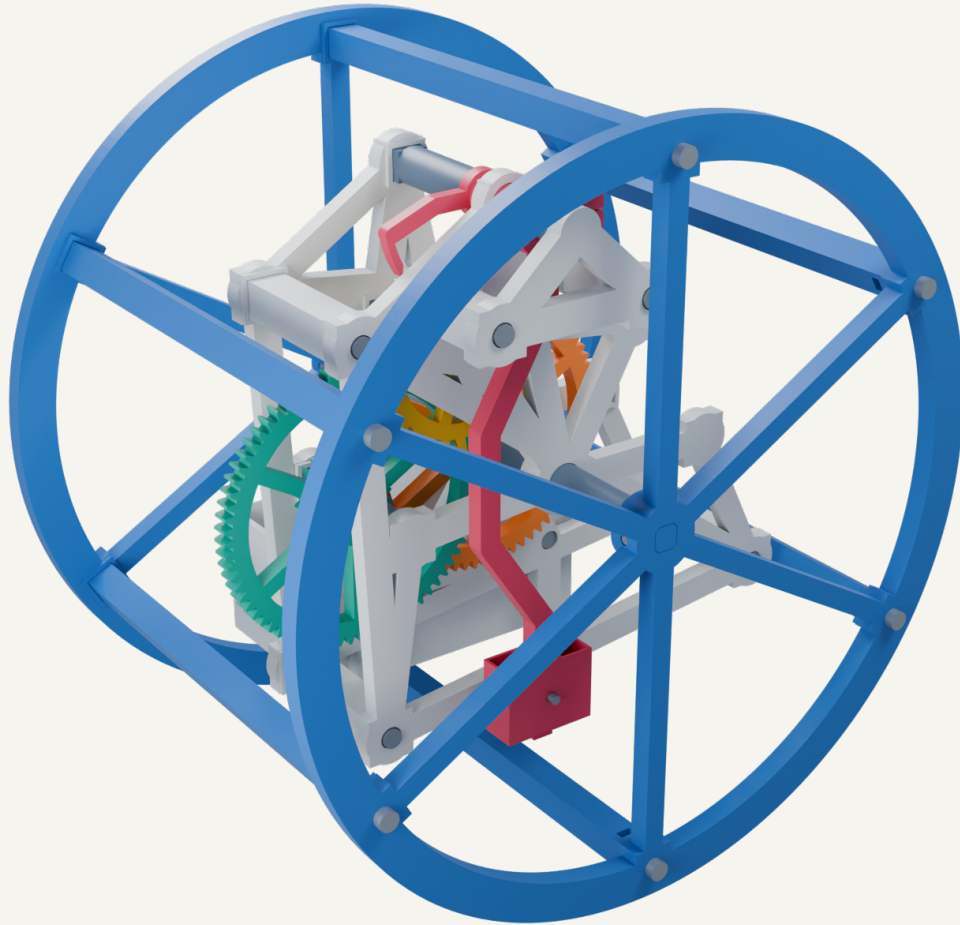
Use the selected folder's B, D and escape compounds, its spacers, and **18_anchor_reversed** from slow-common/. Small pinions face rearward.

D stack: **28 (13.2 mm)**, D compound, **29 (11.7 mm)**. C's rear spacer 17 becomes **25.2 mm**. B spacers and C's 4.7 mm front spacer stay the same lengths.

32x to 64x: swap only D and the escape compound. Both final pairs total 90 teeth, so shaft centres stay 56.25 mm apart.

Use each ratio's print list. The reversed wheel and anchor are a pair: the extra external mesh reverses the escape-wheel direction. Input still turns counterclockwise from the pendulum side. Recheck beat after a swap.

Run on the incline and calibrate



Start with 16x, about 600 g in the main cup and 15-20 g in the bob. Use a straight board with a soft stop at the low end.

The earlier clock ran well with **4 cm rise over 80 cm**, also at 3 cm. At 2 cm it required careful beat adjustment. Begin at 4 cm (about 2.9 degrees).

Time a marked distance. Calculated one-metre times are about **7:42 / 15:25 / 30:50** for 16x / 32x / 64x. R3's changed wheel and pendulum need a new real measurement.

Move the bob lower to slow its rate. Lift the drum to reset uphill. Try 32x only after 16x runs reliably, then try 64x: each slower option leaves less ideal torque at the escapement.

A little rocking can accompany the exchange of angular momentum between pendulum, carrier and drum. Snug joints remove looseness but do not guarantee zero rocking. Check for rubbing, skipped releases or stops; record these separately from timing error.

Keep these lengths and screw counts handy

Part	Position	16x (mm)	32x / 64x (mm)
15	B rear	1.2	1.2
16	B front	23.7	23.7
17	C rear	13.2	25.2
27	C front	4.7	4.7
28 / 29	D rear / front	-	13.2 / 11.7
20	Anchor rear	34.2	34.2
21	Anchor front	7.7	7.7
34	Main front thrust	26.4	26.4
35	Main rear washer	1.2	1.2
36 / 37	Rear / front journal	70 / 45.6	70 / 45.6

One nut per screw	
Case posts	12 x M3x16
All axle retainers	3 x M3x16
Pendulum clamp + bob	2 x M3x16
Frame posts	6 x M3x10
Bridge posts	4 x M3x10
Bowl	2 x M3x10
Gear / anchor feet	3 x M3x10 (16x)
	4 x M3x10 (32x / 64x)
R3 check results: closed connected print meshes; no unintended nominal intersections; spur and pallet checks; -15 to +15 degree pendulum sweep; full drum rotation. Nominal gap to the rotating posts is at least 2.35 mm.	

This guide documents a geometrically checked prototype, not a physically tested R3 clock. Keep the first working version as a reference. Report the selected ratio, slope, ballast, measured run time and any contact or skipped beats.